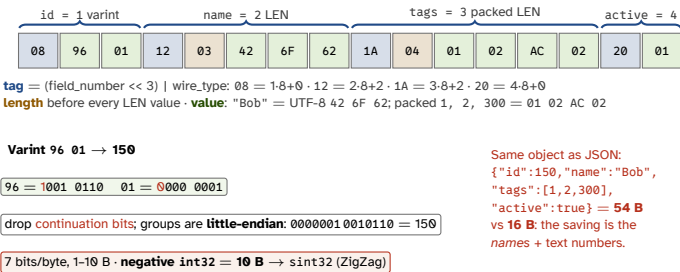
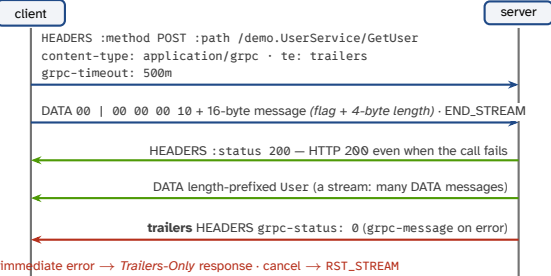


In one line: **Protocol Buffers** is a schema-first binary encoding: every field travels as **(field number, wire type) + value** — names never do, so the **field numbers are the contract**. **gRPC** is RPC over **HTTP/2**: `POST /pkg.Service/Method`, length-prefixed protobuf messages on one stream, the result in **trailers** (`grpc-status`); unary + three streaming kinds, **deadlines** built in.

One User message on the wire — 16 bytes



One unary call = one HTTP/2 stream



Protocol Buffers

- Wire types:** 0 VARINT (int32/64, uint, sint, bool, enum) · 1 I64 (fixed64, double) · 2 LEN (string, bytes, messages, packed repeated) · 5 I32 (fixed32, float); 3/4 = deprecated groups.
- Field numbers** 1–536,870,911 (19000–19999 reserved by protobuf). 1–15 cost a **1-byte** tag, 16–2047 two — give hot fields 1–15.
- Defaults are not sent** (proto3): 0, "", false look exactly like "unset". Need the difference? optional (explicit presence, hasX) or a wrapper message.
- Field order is irrelevant; a non-repeated field seen twice: **last one wins**. **Unknown fields are preserved** and re-serialised (proto3).
- No self-description: without the .proto the bytes are numbers and blobs. Max 2 GiB.

Schema evolution — old apps live for years

- Never change or reuse a field number.** Deleting a field? reserved 3; and reserved "email"; so nobody reuses it.
- Add fields with new numbers:** old clients skip them (unknown), new clients read defaults from old data. Safe type swaps: int32/uint32/int64/uint64/bool.
- Renaming** is wire-safe (names are not sent) but breaks the JSON mapping.
- Enums:** first value 0 = X_UNSPECIFIED; old apps get unknown numbers (Swift: .UNRECOGNIZED(Int)) — handle a default.
- Breaking: singular ↔ repeated, scalar ↔ message. Add a new field / RPC or a pkg.v2 service instead; gate in CI (buf breaking).

Example — contract + Swift client (gRPC Swift 2)

```
syntax = "proto3";
package demo;
message User {
  int32 id = 1;
  string name = 2;
  repeated int32 tags = 3;
  bool active = 4;
  optional string nickname = 6; // presence: hasNickname
  reserved 5; reserved "email"; // deleted in v1.4
}
service UserService {
  rpc GetUser (GetUserRequest) returns (User); // unary
  rpc Watch (WatchRequest) returns (stream Event); // server
  rpc Upload (stream Chunk) returns (UploadResult); // client
  rpc Chat (stream Msg) returns (stream Msg); // bidi
}

try await withGRPCClient(transport: .http2NIOPosix(
  target: .dns(host: "api.example.com", port: 443),
  transportSecurity: .plaintext)) { client in // local dev only
  let users = Demo.UserService.Client(wrapping: client)
  let user = try await users.getUser().with { $0.id = 150 }
}
```

gRPC

- Four kinds** (see the service): each call is one HTTP/2 stream on a shared connection.
- Deadlines:** `grpc-timeout` (units H M S m u n; 500m = 500 ms); propagate the remaining budget downstream. Expiry → `DEADLINE_EXCEEDED`.
- Load balancing:** one long-lived HTTP/2 connection pins every call to one backend behind an L4 balancer — use L7 (Envoy) or client-side balancing.
- Browsers** can't read trailers → **gRPC-Web** (via a proxy) or **Connect** (plain POST, HTTP/1.1 OK, JSON or binary).
- Swift: gRPC Swift 2** (Feb 2025): async/await, Swift 6, pluggable transports (HTTP/2 on SwiftNIO, in-process for tests). **Connect-Swift:** Connect / gRPC-Web / gRPC over URLSession (gRPC needs ConnectNIO).

Status codes you will be asked about

0 OK	1 CANCELLED (caller gave up)
3 INVALID_ARGUMENT (bad input)	4 DEADLINE_EXCEEDED (maybe done!)
5 NOT_FOUND · 6 ALREADY_EXISTS	7 PERMISSION_DENIED
8 RESOURCE_EXHAUSTED (quota)	9 FAILED_PRECONDITION
12 UNIMPLEMENTED (old server!)	13 INTERNAL · 2 UNKNOWN
14 UNAVAILABLE: retry + backoff	16 UNAUTHENTICATED (refresh token)

REST/JSON vs GraphQL vs gRPC

	REST/JSON	GraphQL	gRPC
contract	optional (OpenAPI)	SDL schema	.proto, required
payload	text + names	JSON, per query	binary, numbers
transport	any HTTP	POST one URL	HTTP/2 (Connect: 1.1)
streaming	SSE / WebSocket	subscriptions	4 native kinds
caching	HTTP / CDN for free	client store	none built in
browser	native	native	proxy or Connect
errors	HTTP status	errors[], 200	grpc-status trailer
evolution	versioned URLs	deprecate fields	new field numbers

Traps

- ! Reusing a deleted field number: old apps misread new data — silent corruption.
- ! Checking the HTTP status of a gRPC call — it is 200; read `grpc-status`.
- ! Retrying `DEADLINE_EXCEEDED` blindly — the server may have finished it.

Remember

“Numbers on the wire; never reuse one; status in trailers.”